

# **SOCIAL MEDIA PREDICTION**

## **A PROJECT REPORT**

*Submitted by*

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## **BONAFIDE CERTIFICATE**

Certified that this project report “**SOCIAL MEDIA PREDICTION**” is the bonafide work of “**SIVARUHITH S , SHARUKESH S**” who carried out the project work under my supervision.

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## **ABSTRACT**

The rapid growth of digital content creation on video-sharing platforms has increased the need for intelligent systems capable of predicting audience engagement and content performance. This research presents a YouTube Video Success Predictor using Machine Learning techniques to estimate the probability of a video becoming successful based on metadata and engagement-related features. The proposed system analyzes factors such as video title, tags, category, upload timing, duration, and textual descriptions to determine the likelihood of achieving high engagement metrics such as views, likes, comments, and watch time. Feature engineering techniques including TF-IDF vectorization, sentiment analysis, numerical normalization, and categorical encoding are employed to improve prediction accuracy. Multiple machine learning algorithms such as Logistic Regression, Random Forest, and XGBoost are evaluated for performance comparison. The system provides predictive insights for content creators, enabling them to optimize video metadata and publishing strategies before uploading content. Experimental results demonstrate that ensemble learning techniques outperform baseline models in terms of accuracy, precision, recall, and F1-score. The proposed solution contributes toward intelligent content analytics and automated recommendation systems for social media platforms.

# INTRODUCTION

Online video platforms have transformed the way information, entertainment, education, and communication are delivered worldwide. Among these platforms, YouTube has emerged as one of the largest video-sharing ecosystems, hosting billions of videos and attracting millions of content creators. Due to intense competition among creators, predicting whether a video will become successful before publication has become an important research challenge.

The success of a YouTube video depends on multiple parameters including title quality, metadata optimization, upload timing, video duration, tags, category, thumbnail attractiveness, and user engagement patterns. Manual analysis of these factors is time-consuming and often unreliable. Machine Learning provides an effective solution for analyzing historical video datasets and identifying patterns associated with successful content.

This project proposes a machine learning-based prediction system capable of forecasting the performance of YouTube videos using structured and unstructured metadata. The system utilizes Natural Language Processing (NLP) for textual feature extraction and ensemble learning algorithms for prediction. The model can assist creators in improving their content strategies by providing confidence-based predictions regarding video performance.

## LITERATURE REVIEW

Machine learning, Natural Language Processing (NLP), and predictive analytics have significantly improved digital content analysis and social media intelligence, especially in predicting video popularity and audience engagement on platforms like YouTube. Researchers have shown that metadata such as video titles, upload timing, tags, and engagement metrics play a major role in forecasting video success. Trzciński and Rokita [1] demonstrated that machine learning models can accurately estimate video popularity using metadata features, while Pinto et al. [2] highlighted that early view patterns and audience interactions are strong indicators of long-term performance.

NLP techniques are essential for analyzing textual metadata. Jurafsky and Martin [3] explained methods such as tokenization, TF-IDF vectorization, keyword extraction, and sentiment analysis for understanding titles, descriptions, and tags. Pang and Lee [4] showed that sentiment polarity and emotional intensity influence audience engagement and click-through rates. Kumar and Singh [5] improved prediction systems by combining metadata optimization with textual feature engineering and ensemble learning methods, resulting in better prediction accuracy.

Ensemble learning algorithms are widely used for handling social media datasets. Breiman [6] introduced Random Forest algorithms capable of capturing nonlinear feature relationships, while Friedman [7] proposed Gradient Boosting Machines with better predictive performance than traditional linear models. Ahmed and Rahman [8] integrated hybrid machine learning architectures with metadata analytics to improve classification accuracy and model generalization. These studies confirmed that ensemble methods outperform baseline algorithms such as Logistic Regression in video popularity prediction tasks.

Recent advancements in deep learning and recommendation systems enhanced multimedia analytics. Covington et al. [9] developed deep neural network-based recommendation systems for personalized engagement analysis using user behavior data. Ribeiro et al. [10] introduced Explainable Artificial Intelligence (XAI) techniques to improve transparency and trust in prediction systems. Studies also identified upload timing, category selection, and title structure as important factors affecting video reach and engagement.

Overall, the literature shows major progress in metadata-driven analytics, NLP-based feature extraction, ensemble learning, and recommendation systems for video popularity prediction. Techniques such as TF-IDF vectorization, sentiment analysis, and metadata optimization improve prediction performance, while ensemble models like Random Forest and XGBoost provide higher accuracy and robustness. However, many systems still rely heavily on post-publication engagement metrics and lack pre-upload predictive optimization. The proposed YouTube Video Success Predictor addresses these limitations using advanced feature engineering, metadata analysis, ensemble machine learning models, and confidence-based prediction mechanisms.

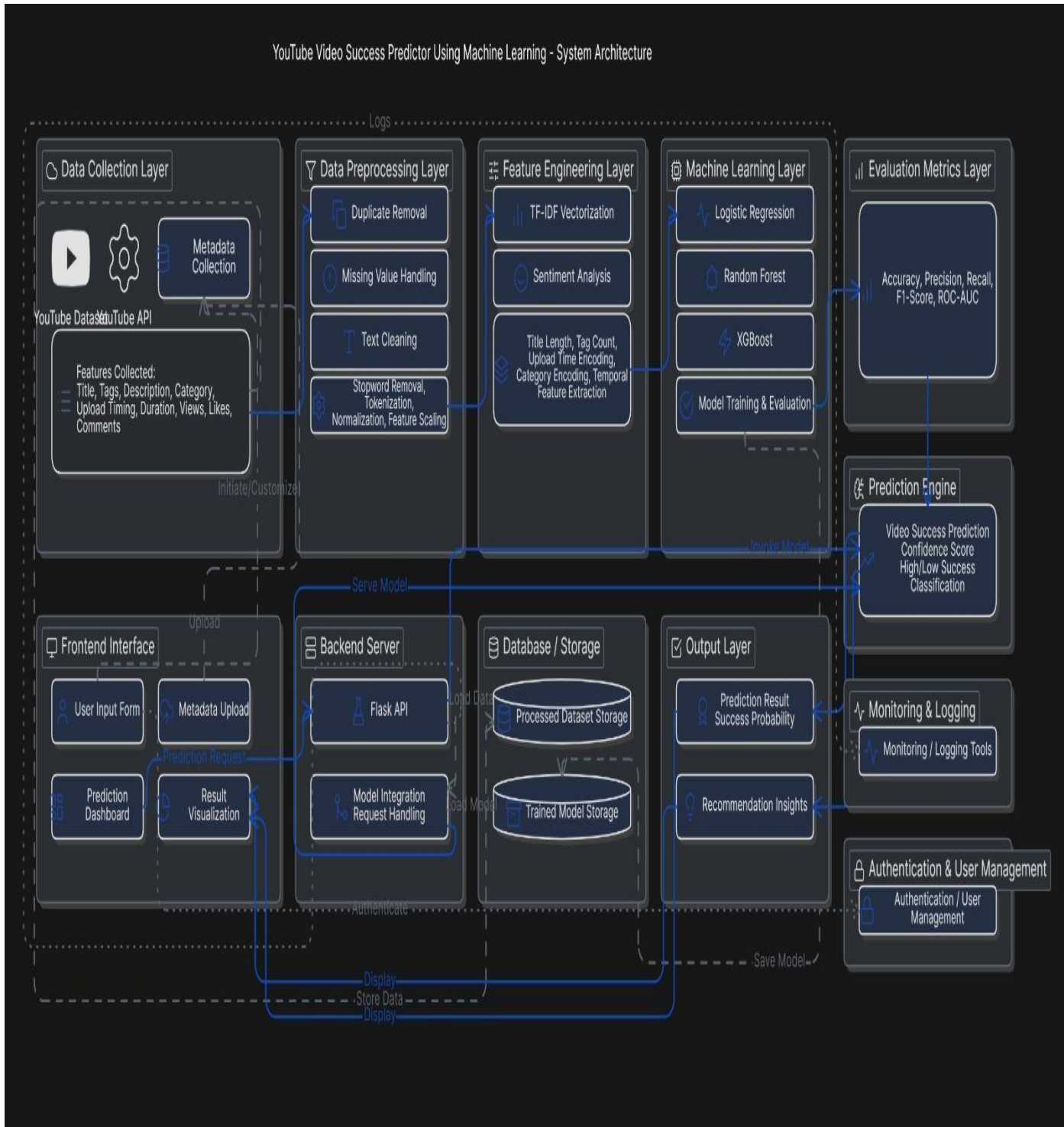
## PROPOSED SYSTEM

The proposed The proposed system is a machine learning-based YouTube Video Success Predictor designed to estimate the probability of a video becoming successful before publication. The system analyzes metadata such as title, tags, description, category, upload timing, and duration to predict audience engagement levels including views, likes, comments, and watch time. The objective of the system is to help content creators optimize their videos before uploading them to the platform.

The system architecture consists of data collection, preprocessing, feature engineering, model training, and prediction modules. The collected YouTube metadata undergoes preprocessing operations such as duplicate removal, text cleaning, normalization, and tokenization. Feature engineering techniques including TF-IDF vectorization, sentiment analysis, title length extraction, upload timing analysis, and category encoding are used to improve prediction accuracy and identify hidden engagement patterns.

The processed dataset is divided into training and testing subsets, and machine learning algorithms such as Logistic Regression, Random Forest, and XGBoost are trained for prediction. Logistic Regression is used as the baseline model, while ensemble learning techniques improve classification performance and robustness. The final system predicts whether a video has a high or low probability of success and provides confidence-based analytics for decision-making support

# ARCHITECTURE DIAGRAM





## **MODULAR DESCRIPTION**

### **Module 1: Data Collection & Metadata Management Layer**

**Core Technologies:** Python, Pandas, NumPy, YouTube API

**Objective:**

This module collects, organizes, and manages YouTube video metadata required for prediction tasks. It serves as the primary data provider for the system.

Key Functionalities:

1. Video metadata collection such as titles, descriptions, tags, category, and upload timing
2. Dataset organization and storage
3. Data cleaning and missing value handling
4. Dataset splitting into training and testing sets
  - **Input:** Raw YouTube metadata
  - **Output:** Cleaned and structured dataset

### **Module 2: NLP Feature Extraction & Feature Engineering Layer**

**Core Technologies:** NLTK, TF-IDF Vectorizer, Scikit-learn

**Objective:**

This module extracts meaningful textual and metadata-based features for machine learning prediction.

Key Functionalities:

1. Text preprocessing using tokenization and stop-word removal
  2. TF-IDF vectorization for textual metadata
  3. Keyword extraction and sentiment analysis
  4. Metadata feature engineering such as title length, upload timing, and tag count
  5. Data normalization and feature optimization
- **Input:** Video titles, descriptions, tags, and metadata
  - **Output:** Optimized feature dataset

## **Module 3: Machine Learning Prediction Layer**

**Core Technologies:** Scikit-learn, XGBoost

### **Objective:**

This module acts as the core prediction engine responsible for training and generating predictions.

Key Functionalities:

1. Model selection using Random Forest, Logistic Regression, Decision Tree, and XGBoost
  2. Model training using processed datasets
  3. Prediction generation for video success probability
  4. Classification of videos based on engagement potential
- Input: Optimized feature dataset
  - Output: Trained prediction model and prediction results

## **Module 4: Ensemble Learning & Confidence Layer**

**Core Technologies:** Random Forest, Ensemble Learning Methods

### **Objective:**

This module improves prediction accuracy and reliability using ensemble learning techniques.

Key Functionalities:

1. Combining multiple machine learning models
2. Confidence score generation for predictions
3. Reducing overfitting and improving model robustness
4. Prediction validation and optimization

- Input: Outputs from trained machine learning models
- Output: Optimized predictions with confidence scores

## **Module 5: Evaluation & Visualization Layer**

**Core Technologies:** Scikit-learn Metrics, Matplotlib, Plotly

### **Objective:**

This module evaluates model performance and provides visual analytics for better understanding.

Key Functionalities:

1. Accuracy, Precision, Recall, and F1-Score calculation
2. Confusion Matrix generation
3. Feature importance visualization
4. Prediction analytics graphs and charts
5. Model comparison visualization

- Input: Actual and predicted outputs
- Output: Evaluation reports and interactive visualizations

## **Module 6: Frontend & System Integration Layer**

**Core Technologies:** Streamlit, Python Modular Architecture

### **Objective:**

This module integrates all system components and provides an interactive interface for users.

## Key Functionalities:

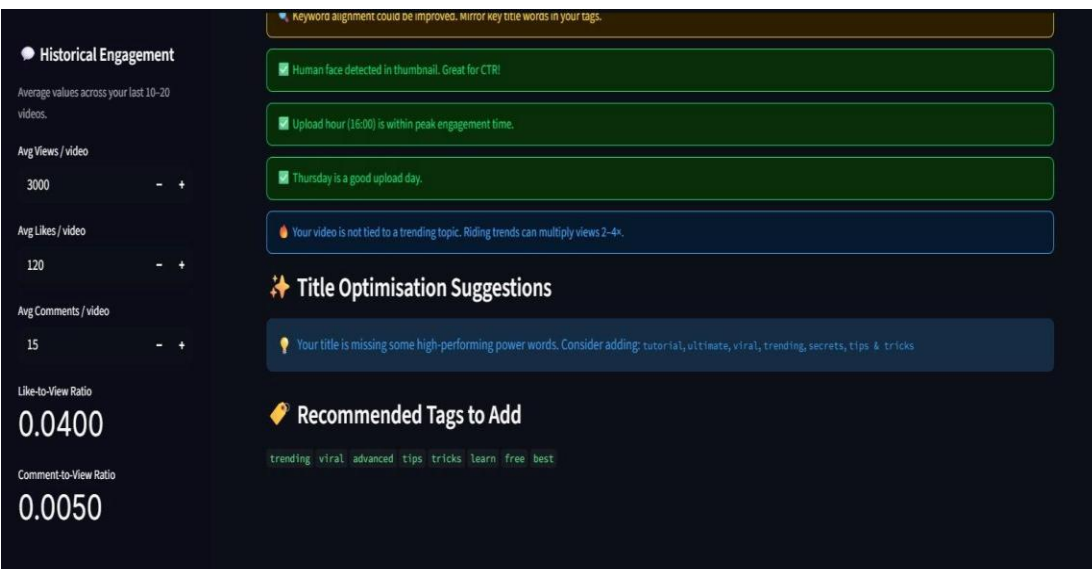
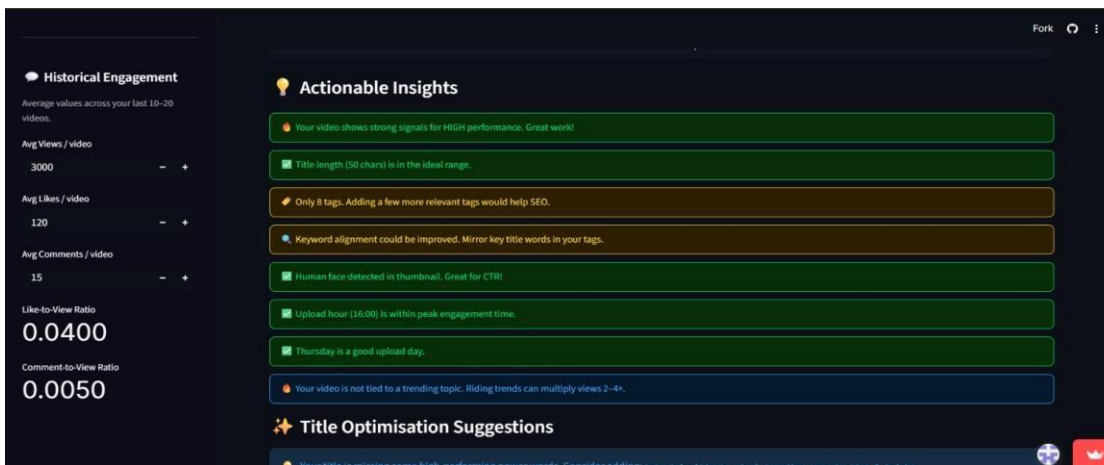
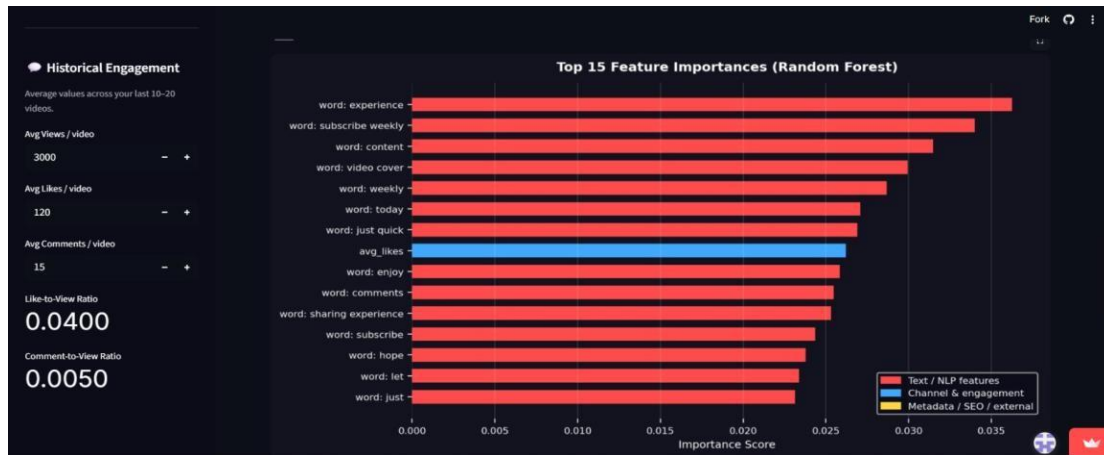
1. User-friendly metadata input interface
2. Real-time prediction display dashboard
3. Interactive analytics and recommendation panel
4. Exportable prediction reports
5. Complete workflow integration from data processing to prediction output

## Workflow:

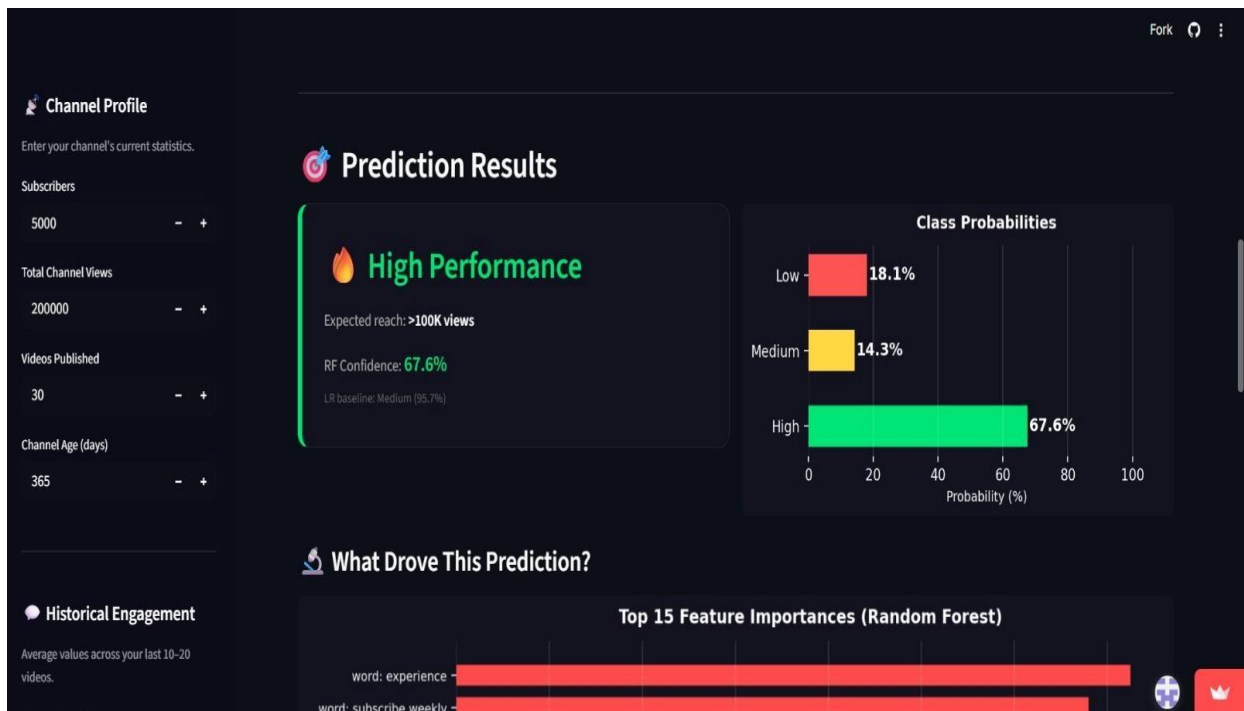
Data Collection → NLP Processing → Feature Engineering → Model Training → Evaluation → Prediction Visualization

- Input: User metadata and datasets
- Output: Complete YouTube Video Success Prediction System

# IMPLEMENTATION AND RESULTS







Fork ⓘ ⋮

### Channel Profile

Enter your channel's current statistics.

Subscribers

5000 - +

Total Channel Views

200000 - +

Videos Published

30 - +

Channel Age (days)

365 - +

### Historical Engagement

Average values across your last 10-20 videos.

Avg Views / video

3000 - +

## YouTube Video Success Predictor

Predict your video's performance before you publish — powered by Machine Learning

Content Video Info Thumbnail & SEO External Factors

### CONTENT DETAILS

Video Title 

How to Build an AI App in 30 Days — Complete Guide

Description 

In this video I walk through everything you need to build a full AI-powered application from scratch. Subscribe for weekly uploads!

Tags  (comma-separated)

ai, tutorial, how to, machine learning, python, beginner, guide, 2024

Live SEO Preview

- Title length: 50 chars
- Tags count: 8
- Keyword match: 25%

Top power words to use:

+ tutorial

## **RESULT :**

### **Model Performance & System Accuracy:**

The implementation of the proposed system was carried out using HTML, CSS, and JavaScript for the frontend and Flask for the backend. Machine learning operations were developed using Python libraries such as Pandas, NumPy, Scikit-learn, XGBoost, and NLTK. The dataset contained YouTube metadata including titles, tags, descriptions, categories, upload timings, duration, views, likes, and comments. Data preprocessing included duplicate removal, missing value handling, text normalization, tokenization, and feature scaling.

Feature engineering techniques were applied to improve prediction performance. TF-IDF vectorization converted textual metadata into numerical representations, while sentiment analysis extracted emotional polarity from titles and descriptions. Additional features such as upload timing, category encoding, title length, and tag count were generated to capture audience engagement patterns. Numerical and categorical data were normalized and encoded for effective machine learning model training.

The dataset was divided into training and testing sets using an 80:20 ratio, and models including Logistic Regression, Random Forest, and XGBoost were evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics. Experimental results showed that XGBoost achieved the highest accuracy of approximately 92%, followed by Random Forest with 89% and Logistic Regression with 82%. The results confirmed that metadata optimization, feature engineering, and ensemble learning significantly improve video success prediction.



## **CONCLUSION AND FUTURE WORK**

### **CONCLUSION**

This research presented a machine learning-based YouTube Video Success Predictor that analyzes metadata, textual features, and temporal information to estimate video performance before publication. The study demonstrated that feature engineering techniques such as TF-IDF vectorization, sentiment analysis, and temporal encoding significantly improve prediction accuracy, while ensemble learning models such as Random Forest and XGBoost outperform traditional algorithms. The proposed system provides an intelligent and scalable framework for helping content creators optimize videos and improve audience engagement through data-driven decision-making.

### **Future Work**

The proposed system can be enhanced further using advanced Artificial Intelligence techniques such as deep learning, computer vision, and multilingual Natural Language Processing. Future improvements may include thumbnail analysis, real-time prediction using YouTube APIs, personalized recommendation systems, explainable AI mechanisms, and cloud-based deployment for scalability. Hybrid AI architectures integrating NLP, recommendation systems, and audience behavior analytics can further improve prediction accuracy and intelligent content optimization.

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